

COURSE DESCRIPTION

Modbus is the most widely deployed fieldbus protocol in industrial automation, present in virtually every manufacturing and utility installation worldwide. This course covers RTU mode on RS-485, ASCII mode, and Modbus TCP over Ethernet, along with the register data model, function codes, exception handling, and data type encoding across vendor implementations. Security limitations and network-layer compensating controls are addressed in the context of modern OT network architecture.

PREREQUISITES

Basic TCP/IP networking concepts. No prior SCADA or industrial protocol experience required.

COURSE TOPICS

1. **Introduction:** Protocol origins, the 1979 register model, and why the original design persists unchanged in modern field devices.
2. **Data Model:** Coils, discrete inputs, input registers, and holding registers — table structure, data types, and access rules.
3. **RTU Mode:** RS-485 framing, CRC-16 integrity, multi-drop bus wiring, and inter-frame silence on serial Modbus links.
4. **ASCII Mode:** Hex-encoded framing, colon delimiter, overhead comparison to RTU, and legacy HMI equipment identification.
5. **Modbus TCP:** Port 502, the MBAP header, transaction ID pipelining, and security implications of Ethernet transport without authentication.
6. **Function Codes:** Read, write, and diagnostic codes; the subset covering ninety-five percent of deployments; error response handling.
7. **Exception Codes:** Eight exception identifiers, meanings, and systematic diagnosis from exception response to root cause.
8. **Data Types:** Register-pair conventions, byte order, word order, and floating-point encoding across vendor implementations.
9. **Troubleshooting:** Addressing offset errors, byte-order mismatches, RS-485 termination faults, firewall drops, and poll rate overruns.
10. **Security:** Protocol-level limitations and compensating controls: network segmentation, unidirectional gateways, and anomaly detection.
11. **Practice Lab:** Register mapping, Exception Code 02 tracing, serial-to-TCP gateway configuration, byte-order diagnosis, and Wireshark traffic analysis.

LEARNING OUTCOMES

- Configure Modbus RTU and TCP device connections from device datasheet parameters.
- Read and interpret function code and exception code responses in a traffic capture.
- Apply byte-order and register-pair conventions to map floating-point and 32-bit values.
- Diagnose serial and TCP-layer Modbus faults using systematic isolation procedures.
- Specify network-layer compensating controls appropriate for a Modbus deployment.

REQUIRED SOFTWARE

- **ModRSim2** (free, Windows) — virtual Modbus RTU/TCP slave for register read/write exercises.
- **Modbus Poll** (trial, Windows) — master client for polling registers and inspecting exception responses.
- **Wireshark** (free, cross-platform) — packet capture for Modbus TCP traffic analysis.